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Here Come the Suns: Stripped Binary Populations in Early Galaxy Formation

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ABSTRACT

With the advent of the James Webb Space Telescope's (JWST) spectroscopy, we can measure the ultraviolet (UV) continuum and the nebular emission lines powered by ionizing radiation at redshifts up to $z \sim 14$. Interpreting these JWST observations at high redshifts requires understanding the spectral energy distribution (SED) of galaxies. Most models used to study galaxy formation derive the SED from single, non-interacting stars, neglecting the role of binary interactions. In close binaries, one star can lose its hydrogen-rich envelope to a companion, leaving behind a hot, stripped helium star. Without their hydrogen envelope, these stripped stars are extremely hot and emit hydrogen- (H) and helium II- (He II) ionizing photons long after the most massive stars have died.

In this work, I will add stripped binary populations to a semi-analytical model called `DELPHI`, reconstruct the star-formation history of every `DELPHI` galaxy with a stripped binary population, and study the effects of stripped binaries on the SED of early galaxies. From the resulting SEDs, I will quantify the effects of stripped binaries on the UV luminosity function (UV LF), the UV luminosity density (UV LD), the UV slope (β), the production rates of H- and He II-ionizing photons, and the ionization efficiency ξ_{ion}.

Without stripped stars, single-star models can underestimate the He II-ionizing output of galaxies in which star formation has ceased. The He II emission is attributed to other sources like active galactic nuclei (AGN) or very massive stars. By including stripped binaries in the star formation histories of `DELPHI` galaxies and by comparing with JWST observations, can we understand how much of the He II emission in early galaxies comes from stripped binaries?

1. INTRODUCTION

The James Webb Space Telescope (JWST) has revolutionized the field of high-redshift galaxy formation by offering groundbreaking sensitivity, spatial resolution, and wavelength coverage. It has enabled astronomers to identify hundreds of star-forming galaxies at redshifts up to $z \sim 14$, rattling the scientific community with two interesting observations: an unexpected abundance of super-early, bright galaxies at the bright end ($M_{UV} \lesssim -21$) of the ultraviolet luminosity function (UV LF); and a growing number of sources that show hard ionizing spectra with strong He II emission. Both observations constrain the intrinsic stellar SED of early galaxies.

To model and predict early galaxy formation, one must model the growth of structure in the early Universe. Fortunately, the cosmological principle suggests that on large scales, the early Universe is homogeneous and isotropic, and the first bound structures are low-mass dark matter halos that collapse into overdense regions. Due to their shallow potential wells and low masses, even a small amount of star formation can provide the necessary feedback to expel all star-forming gas from these systems. To resume star formation, a low-mass halo must wait for gas to be accreted from the surrounding intergalactic medium (IGM) or to be deposited via mergers. Dayal et al. (2014) developed a semi-analytical model called `DELPHI` that embeds this simple principle: at any given time, there exists a limiting star formation efficiency such that the energy produced by newly formed stars is sufficient to expel all remaining gas, at least temporarily.

In most models like `DELPHI` , it is assumed that stellar populations consist of single stars. `DELPHI` uses the single-star `Starburs99` models (Dayal et al. 2014; Mauerhofer et al. 2025). At each time step, single-star population-synthesis tables are used to evaluate the light of a stellar population of a given age and metallicity. Therefore, for a chosen initial mass function, that table fixes the 1500 Å luminosity and the production rates of H- ($E_\gamma > 13.6$ eV) and He II-ionizing ($E_\gamma > 54.4$ eV) photons. However, binary interactions via mass transfer open a new possibility.

For instance, a stripped star is a hot, helium-rich core left after a binary companion removed its hydrogen envelope. These stars emit many photons with $E_\gamma >$ 13.6 eV (hydrogen-ionizing) and even $E_\gamma > 54.4$ eV

(helium-ionizing). With the inclusion of stripped stars, we expect the following changes in our understanding of early galaxies:

- Stripped stars emit most of their ionizing photons after massive stars have died. They don't appear until 10 Myr after a burst and dominate the amount of ionizing photons output until at least 100 Myr (Götberg et al. 2019). As a consequence, single stars may underestimate the H- and He II-ionizing photons produced in galaxies whre star formation stopped more than about 10 Myr ago.

- Stripped stars add very little light at 1500Å (the non-ionizing UV light) unless star formation stopped more than ~ 500 Myrs ago (Götberg et al. 2019). Thus, we expect the UV LF and UV LD predicted by `DELPHI` with co-eval single-star populations not to change much. However, we still expect a change in ξ_{ion}.

- A nebular He II emission is one of the best indicators for the first, metal-free (Population III or Pop III) stars. Stripped stars in ordinary, metal-enriched populations also produce He II-ionizing photons. If Pop III stars and stripped star populations both produce He II, can we use their emission-line strengths to prevent metal-enriched populations (Population II or Pop II) with stripped binaries from masquerading as Pop III stars?

This work aims to add the stripped binary population to DELPHI and determine how it changes the SED of early galaxies, from the UV to the He II-ionizing range.

2. SED OF AN EARLY GALAXY

We can divide the SED into three separate regions for the purposes of this work:

- **UV - non-ionizing**. $E_\gamma < 13.6\,eV$. We can understand how photons behave in this region by directly observing the UV luminosity function, the UV luminosity density and the UV slope β.

- **H-ionizing** $E_\gamma > 13.6\,eV$. These photons are absorbed by the gas and seen through hydrogen recombination lines (Lyα, Hα, Hβ). We can use this region to find the photon rate $Q(H)$ and $\xi_{ion} = Q(H)/L_{UV}$.

- **He II-ionizing** $E_\gamma > 54.4\,eV$. These photons ionize He^+. They are seen through He II λ1640 and λ4686. We can use this region to find the photon rate Q(He II) and the hardness $Q(HeII)/Q(H)$.

Stripped stars should add only a very small fraction to the UV window, while massive stars of a population are alive. Their contribution grows towards higher photon energies and with time once massive stars cease to exist. Their ionizing emission is harder than that of massive stars, and they dominate the ionizing output of populations a few Myrs after star formation stops.

In Figure 1, I use the data from `Starburst99` and Götberg et al. (2019) to reproduce Figure 7 of their paper. In Figure 1, I show the ionizing spectrum for a population formed in a single instantaneous bust (co-eval stellar population) at four metllacities. From Figure 1, it is evident that stripped stars harden (a harder spectrum means there are more high-energy photons relative to low-energy photons) the ionizing spectrum. The spectra that includes stripped stars ($11 - 100$ Myr) falls off more slowly in energy compared to the single-star spectra ($2 - 10$ Myr).

A `DELPHI` galaxy is not a single burst but sum of multiple bursts. My project wants to sum up the `DELPHI` SEDs for multiple bursts and for each galaxy, produce an SED comparable to Figure 1 with and without stripped binaries.

2.1. *Star Formation History to a SED*

The SED of a galaxy observed at a time t_{obs} is the light of all stars it has formed and can be given by this equation (Conroy 2013):

$$SED(\lambda) = \int_0^{t_{obs}} SFR(t) l_\lambda(t_{obs} - t, Z(t)) dt \quad (1)$$

where $SFR(t)$ contains the star formation history of the galaxy, $Z(t)$ is the metallicity of the gas the stars formed from and $l_\lambda(t, Z)$ is the light per unit stellar mass of a population of age t and metallicity Z (summed up over the IMF). In `DELPHI` , we use the Krouppa IMF and the SFR is constant within 30 Myr steps, so the integral becomes a discrete sum over steps. Hence, the stellar SED of a galaxy in `DELPHI` will be built from its star-formation history in three steps (Figure 2):

- `DELPHI` stores, in 30 Myr steps, the stellar mass formed in each halo $M_*^{new} = SFR \times \Delta t$, and the metallicity of the gas it formed from. SFR, here, is the star formation rate.

- At each burst, we can find the SED, per unit SFR, of stars formed at a constant rate during one step and at that age. We find the luminosity per unit stellar mass for single stars (from Starburst99) and stripped stars: l_λ^{single} and l_λ^{strip} respectively.

- Sum over all bursts k, where burst k formed n_k steps before galaxy j is observed, at the rate SFR_k

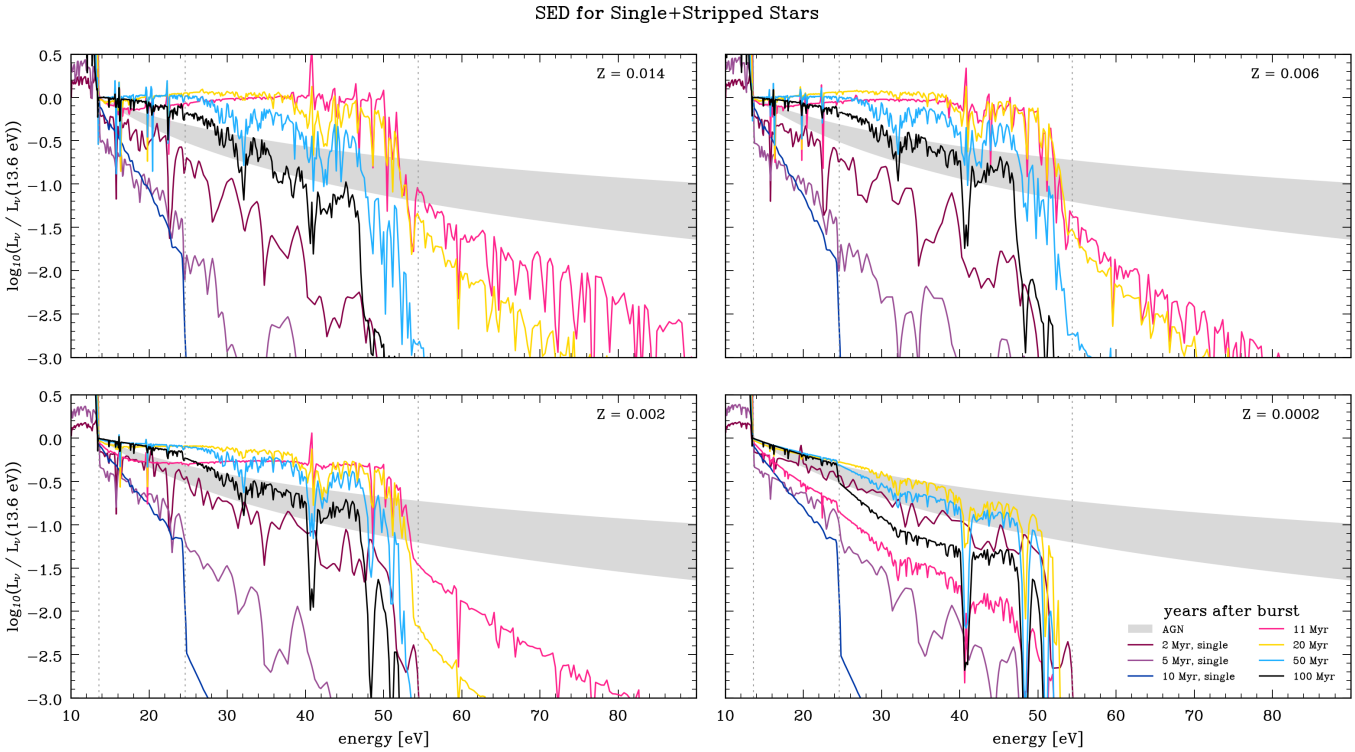


Figure 1. The ionization spectra of co-eval stellar populations for four metallicities: Z= 0.014, 0.006, 0.002, 0.02. They are made from the data published by Götberg et al. (2019). Single stars are shown 2, 5, and 10 Myr after the burst. Single+stripped stars are shown at 11, 20, 50 and 100 Myr. The gray band is accounting for the AGN with slopes $L_\nu \propto \nu^{-1.2} - \nu^{-2}$. The dotted, vertical lines are teh energies at 13.6, 24.6 and 54.4 eV.

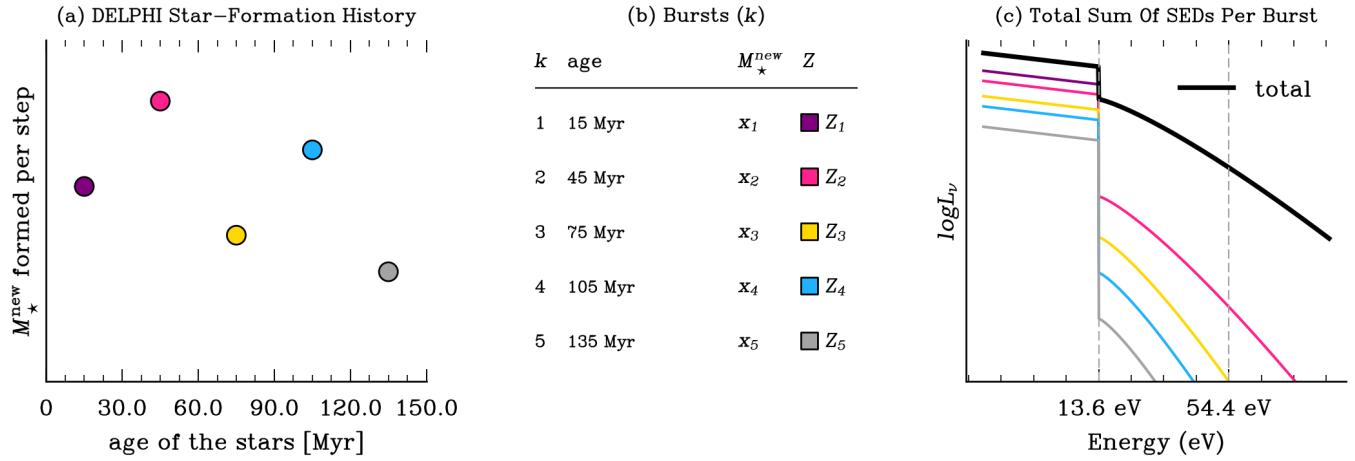


Figure 2. We show how SEDs are built in `DELPHI` through this illustration. *Left/Panel (a)* The stellar mass formed in each 30 Myr step inside `DELPHI` is labeled as $x_1, x_2, ..., x_5$. The colors represent the metallicity of the gas. *Center/Panel (b)* This shows different rows for different bursts (k). For each burst, we see the age, stellar mass formed and metallicity. *Right/ Panel (c)* This shows the SED of each burst, summed up to give the total SED of the galaxy in black. The vertical lines are the energies at 13.6 eV and 54.4 eV

and metallicity Z_k. The sum is the total SED given by:

$$SED(\lambda, j) = \sum_k SFR_k \Delta t [l_\lambda^{single}(n_k, Z_k) + l_\lambda^{strip}(n_k, Z_k)] \quad (2)$$

where $SFR_k \Delta t = M_*^{new}$ is the stellar mass formed in burst k and $l_\lambda(n_k, Z_k)$ is the l_λ from Equation 1 averaged over the age of its stars, from $(n_k - 1)\Delta t$ to $(n_k)\Delta t$

The first stripped stars appear around 10 Myr after a burst (Götberg et al. 2019). The stripped stars exist between 10 Myr to 2 Gyr. Since `DELPHI` follows each burst up to 890 Myr, stripped stars can contribute ionizing photons to the bursts in panel(b) of Figure 2. In panel (c) of Figure 2, I have visualized the SED of each individual bursts without stripped binaries, and added them up to give the total SED. Once I include stripped binaries to the SED, I expect each of these older bursts to gain a trail and harden the total SED. Furthermore, once I can find the SED, I can investigate Q(H), Q(He II) and ξ_{ion} to see how they change.

2.2. *Where do He-II ionizing photons come from?*

In single-star Pop II models, photons above 54.4 eV are rare, so nebular He II in a galaxy with a Pop II population is usually attributed to sources like AGN, accreting black holes, X-ray binaries (Schaerer et al. 2019), or, at high redshift, Pop III stars. Even in metal-poor star-forming galaxies, He II is often stronger than single-star population models predict (Shirazi and Brinchmann 2012; Senchyna et al. 2017). Stripped binaries are missing from single-star models, yet they emit He II-ionizing photons (above 54.4 eV in Figure 1). An outcome of this work is to answer whether some of the He II that is currently thought to originate from AGN, black holes or Pop III stars could actually be from stripped stars.

2.3. *Impact of Stripped Binaries On Search For The First Stars*

Hard ionizing radiation, traced by nebular He II emission, is one of the best signatures of Pop III stars (Venditti et al. 2026). Figure 3 shows why this signature is difficult to use in practice. A Pop III-hosting galaxy passes through a pristine phase, a self-polluted phase (where it is pure Pop III), and a hybrid phase in which a Pop II population forms alongside the surviving Pop III stars. The Pop III-dominated phase($M_{\star,PopIII} > 0.5 M_{\star,tot}$) lasts only $\sim$ 15-20 Myr.

In Figure 3, on the EW(He IIλ4686) versus [OIII]λ5007/$H\beta$ panel, the self-polluted and Pop III-rich galaxies lie above the lower limit of the Pop III threshold established in Nakajima and Maiolino (2022). Pop III-mid galaxies spread across the threshold, and Pop III-poor galaxies fall well below it, close to the region where AGN and other sources like black holes produce similar line ratios. Stripped helium stars are among these contaminants (Venditti et al. 2026), and they are one that I will attempt to make `DELPHI` model.

Stripped stars in the Pop II component of a hybrid galaxy can raise EW(He II) and so create false Pop III candidates. In a hybrid galaxy, the Pop II component contains binaries whose stripped stars keep emitting He II ionizing photons after its massive stars have died (Götberg et al. 2019). This can raise EW(He II) without any Pop III stars and push Pop III-poor or Pop II-only galaxies towards the Pop III threshold, creating false Pop III candidates.

This contamination should be strongest in enriched gas, and should depend strongly on metallicity since the temperature of the stripped stars falls at low metallicity. At Z$\sim$0.0002, binary stripping leaves a thin hydrogen layer on the star, making it cooler, so its He II-ionizing output is about 100 times lower than at higher metallicity (Götberg et al. 2019). In addition, massive low-metallicity stars do not expand enough to be stripped, so the most massive and hottest stripped stars do not form (Hovis-Afflerbach et al. 2025) Modeling stripped stars in `DELPHI` yields the He II emission that Pop II populations alone can produce as a function of metallicity and redshift. This new boundary or floor will help determine whether a He II detection is appropriate to strengthen the case for a Pop III star or not. A He II detection above the floor is a Pop III star, while a detection below it is a Pop II contaminant masquerading as Pop III.

3. MILESTONES AND TIMELINE

I propose the following between the months of October to May for my AST1501Y project, supervised by Prof. Pratika Dayal. In my timeline, I have taken into account the time required to learn `DELPHI`, meet 1501Y deadlines and conduct the research. A gantt chart to summarize this timeline is shown in Figure 4.

Note that a buffer time for the minor existential crises expected around the holidays was not included in the timeline.

- **October** - I have already learned DELPHI, compiled it, and run it with five seeds for the fiducial models on the 600 input merger trees. I have reproduced the UV luminosity function, UV luminosity density, halo mass function and stellar mass function plots from Mauerhofer et al. (2025).

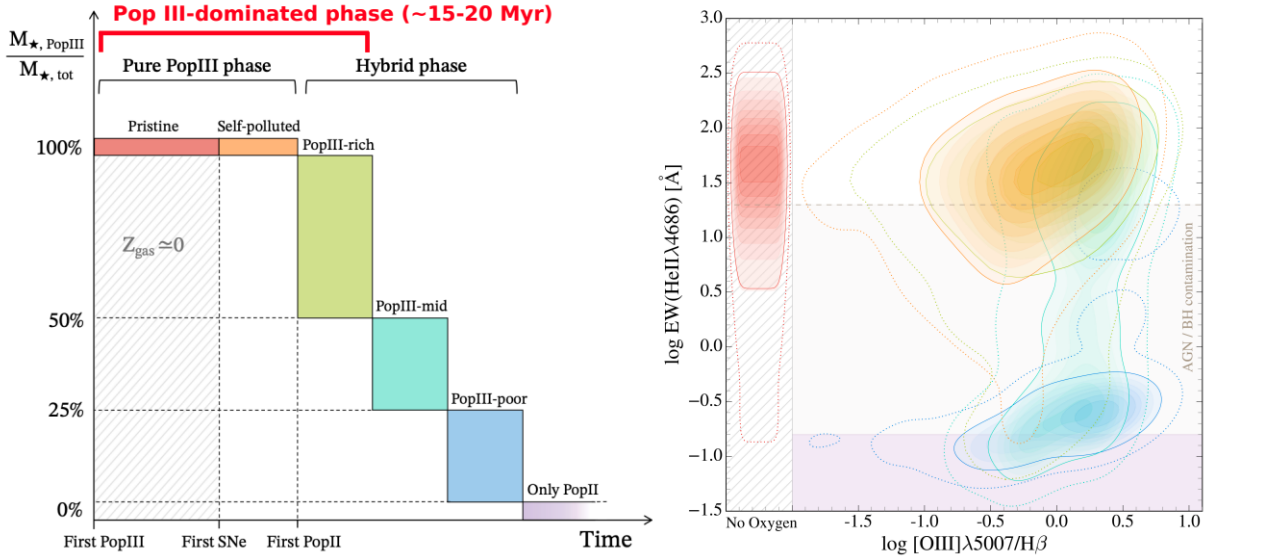


Figure 3. Fig. 10 of Venditti et al. (2026), adapted from Rusta et al. (2025). *Left:* evolutionary stages of a Pop III-hosting galaxy, ordered by the Pop III fraction of its stellar mass. The Pop III-dominated phase ($M_{\star,PopIII} > 0.5\,M_{\star,tot}$) lasts ∼15-20 Myr. *Right:* Distribution of Pop III-hosting galaxies predicted by the `NEFERTITI` model on the EW(He IIλ4686) versus [OIII] λ5007/Hβ diagram. Colors match the stages on the left; solid (dotted) contours enclose 68% (95%) of the galaxies. The dashed line marks the Pop III threshold of Nakajima and Maiolino (2022), and the lilac region below it the zone contaminated by AGN and direct-collapse black holes.

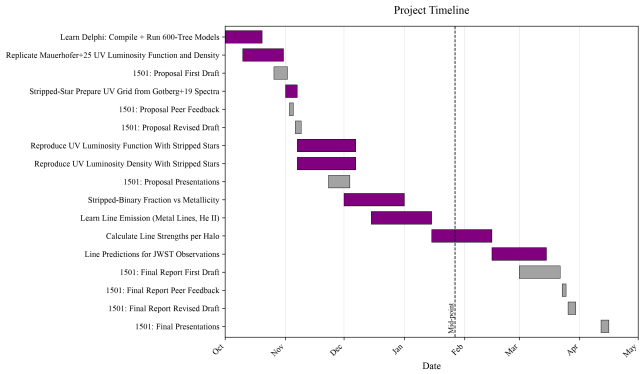


Figure 4. Gantt Chart Of The Project Timeline. In gray, I have the requirements needed by the course. In purple, I have the research goals I have to meet.

- **November** - I will add the stripped-star UV grid and calculate the UV results. During this period, I will also work on the proposal, its peer feedback (and giving feedback), and the presentation.
- **December** - I will finish the integration of stripped stars and their UV spectra from Götberg et al. (2019) into `DELPHI`. I will then download the catalog from Hovis-Afflerbach et al. (2025) to find the fraction of stripped-binary as a function of metallicity.
- **January** - I will start reviewing line strengths, finish the stripped-binary fraction plot against metallicity, decide which lines to model (e.g., $H\alpha, H\beta$ and how to treat escape and ionization parameters. I will also prepare the 1501Y midpoint update.
- **February** - I will finish the line strengths for each halo with and without stripped stars, and turn them into observable predictions for JWST (equivalent widths, line ratios, line luminosity functions) at the redshifts used for the UV results.
- **March** - I will concentrate on finishing the JWST line predictions this month and start writing the final 1501 report.
- **April** - I will prepare and give the 1501 final presentation.

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